AWS Compute

SAA-C03 exam-ready reference

Aligned to current AWS terminology. This version removes vague shortcuts and keeps only the rules that stay true on the exam.

Use this guide to decide, not to memorize slogans. The exam rewards service selection based on constraints, not analogies.

- Know the decision rule for EC2 families, pricing, storage, load balancing, Lambda, containers, and Beanstalk.
- Prefer the service that matches the workload constraint: latency, cost, durability, scaling, or operational overhead.
- Treat the trap boxes as the exam shortcut: they tell you what the wrong answer is trying to exploit.

1. EC2 Instance Families

Choose the family from the workload shape, then confirm the generation and suffix. The family is usually the real answer on the exam.

Family	Best fit	Typical examples	Exam cue
General purpose	Balanced CPU and memory	M / T families	Web apps, app servers, default choice
Compute optimized	High CPU per dollar	C families	Batch, HPC, video encoding, CPU-heavy
Memory optimized	Large working set in RAM	R / X / U families	In-memory DB, SAP HANA, caching
Storage optimized	Fast local or dense disk	I / D / H families	NoSQL, log processing, big I/O
Accelerated computing	GPU, ML, inference	P / G / Trn / Inf families	AI/ML, CUDA, graphics, training

Key nuance: T-family instances are burstable. They accumulate CPU credits while idle and spend them while bursting. If credits run out, CPU drops to baseline. For sustained CPU load, the right answer is usually a non-burstable M or C family.

Suffix	Meaning	Exam use
g	Graviton/ARM	Better price/performance when the app supports ARM
a	AMD	Common cost-optimized alternative
i	Intel	Standard compatibility or specific Intel requirement
d	Instance store	Local disk attached to the host
n	High network	Bandwidth-heavy workloads

2. EC2 Pricing Models

Use the commitment level and interruption risk to decide. Savings Plans and Spot are the two words that most often show up in cost questions.

Model	What you commit to	Best for	Interruptible?
On-Demand	Nothing	Unpredictable or short-lived workloads	No
Savings Plans	Hourly spend commitment	Flexible steady usage; Compute SP covers EC2, Fargate, Lambda	No
Reserved Instances	Specific capacity pattern	Predictable EC2 usage, older exam wording	No
Spot	No upfront commitment	Fault-tolerant, batch, interruption-tolerant jobs	Yes
Dedicated Instances / Hosts	Physical isolation	Licensing, compliance, or host-level control	No
Capacity Reservations	Capacity only	Need guaranteed capacity in an AZ	No

Decision rule: if the question says save money without losing flexibility, start with Savings Plans. If it says spare capacity and interruptions are acceptable, use Spot. If it says you need reserved capacity in a specific AZ, look for Capacity Reservations.

Exam trap: Reserved Instances are a billing construct, not a capacity guarantee. Capacity Reservations guarantee capacity; Savings Plans lower the cost of usage.

3. Storage

The exam usually reduces storage to three choices: local ephemeral disk, block storage, or shared file storage.

Storage	Durability	Scope	Good answer when the question says...
Instance Store	Ephemeral	Host local	Need the fastest local scratch disk; data can disappear on stop/terminate
EBS	Persistent block	Single AZ	Need boot volume, transactional block storage, or volume snapshots
EFS	Persistent file	Multi-AZ regional	Many EC2 instances need the same shared files at once

EBS type	Use when
gp3	General purpose baseline with tunable performance
io2	Very high IOPS, mission-critical databases
st1	Throughput-heavy, sequential HDD access
sc1	Cold, infrequent access on HDD

Important fact: EBS volumes live in one Availability Zone and must be attached to an instance in the same AZ. EFS Regional file systems can be mounted from instances in multiple AZs.

4. Auto Scaling Groups

Use ASG when the workload must scale horizontally or self-heal. The exam usually wants the policy type or the health-check behavior.

Term	Meaning	Exam cue
Min	Lowest desired fleet size	Always keep at least this many instances
Desired	Current target fleet size	Actual target count right now
Max	Upper bound	Never exceed this many instances
Target tracking	Maintain a metric	Keep CPU around a target percentage
Step scaling	Scale by breach size	Different alarm sizes, different scale steps
Scheduled	Scale on a timetable	Predictable traffic peaks
Health checks	Replace bad instances	Use EC2 and ELB health checks together

Exam cue: if the workload needs time to initialize, look for launch warm-up / grace period / warm pool language. If the question says the fleet should react to CPU or queue depth automatically, target tracking is usually the best answer.

5. Load Balancers

Pick the balancer by layer and by what the exam is telling you to preserve: host/path routing, static IPs, or inline appliance chaining.

Type	Layer	Best for	Key clue
ALB	Layer 7	HTTP/HTTPS, path or host routing, WAF, microservices	Rules use host-header, path-pattern, query-string, source-ip
NLB	Layer 4	TCP/UDP/TLS/QUIC, static IP, extreme performance	Static IPs, source IP preservation, high throughput
GWLB	Layer 3	Security appliances and service insertion	GENEVE on port 6081
CLB	Legacy	Only if the question is old or says classic	Usually not the answer

ALB rule cue: host-based routing and path-based routing belong to ALB. If the problem is about HTTP request content, ALB is usually the right load balancer. If it is about preserving client IP or handling TCP/UDP, think NLB.

6. AMIs and Placement Groups

AMI is the image. Placement groups are about where the instances land. The exam often swaps those two on purpose.

Concept	What it is	Important exam fact
AMI	Reusable template for EC2 instances	Region-specific; copy it to another Region if needed
EBS-backed AMI	Snapshot-backed image	Can be copied across Regions and accounts
Cluster placement	Close together in one AZ	Lowest latency, highest throughput, great for HPC
Spread placement	Separate hardware	Use for a small number of critical instances
Partition placement	Separate racks per partition	Use for large distributed systems like Hadoop/Cassandra/Kafka

Placement rule: cluster is for low latency, spread is for fault isolation, partition is for large distributed systems. If the question says 'maximum resilience for a few critical instances,' spread is the answer. If it says 'low-latency node-to-node traffic,' choose cluster.

7. Lambda

Lambda is the right answer when you want event-driven compute with no server management and short execution windows.

Fact	Exam-safe value
Memory range	128 MB to 10,240 MB
Timeout	Up to 15 minutes
Ephemeral storage	512 MB to 10,240 MB in /tmp
Scaling	Automatic per event / request
Cold-start mitigation	Provisioned concurrency
Best use	Event processing, automation, lightweight APIs, scheduled jobs

Decision rule: if the job is short, stateless, and event driven, Lambda is usually the best answer. If the job needs long runtime, large local scratch space, or you want to keep containers warm, use ECS/Fargate or EC2 instead.

8. Containers and Elastic Beanstalk

The exam usually wants you to distinguish orchestrator, infrastructure model, and deployment abstraction.

Service	What AWS manages	When to pick it
ECS	Container orchestration	AWS-native container platform with low operational overhead
EKS	Managed Kubernetes control plane	You need Kubernetes APIs and ecosystem compatibility
Fargate	No servers or node groups	Serverless containers for ECS or EKS
Elastic Beanstalk	Deployment orchestration on top of EC2 / ALB / ASG	You want app deployment without building the platform yourself

Exam trap: Beanstalk is not the same thing as Lambda, and Fargate is not Kubernetes itself. Fargate is the serverless compute layer for containers. EKS is Kubernetes. ECS is AWS-native orchestration.

9. Master Keyword Map

Use this as the fastest translation layer on the exam. Read the clue, then jump straight to the service.

If the question says...	Think...
host-based or path-based HTTP routing	ALB
static IPs, TCP/UDP/TLS, high throughput	NLB
service insertion / firewall appliance	GWLB
shared files across many EC2 instances	EFS
fast local scratch disk	Instance Store
boot volume, snapshots, block storage	EBS
event-driven code with no servers	Lambda
Kubernetes requirement	EKS
AWS-native container orchestration	ECS
simple deploy pipeline for web apps	Elastic Beanstalk

10. Top Exam Traps

T3 trap: T instances are burstable. If the load is sustained, CPU credits run out and performance drops. Pick M or C for steady CPU.

EBS trap: EBS is AZ-scoped and persistent. It is not the answer when the question asks for multi-AZ shared file access.

ALB vs NLB trap: ALB is for HTTP/HTTPS routing decisions. NLB is for TCP/UDP/TLS and static IP needs.

Placement trap: Cluster gives low latency, not high availability. Spread gives failure isolation, not high fan-out density.